

To combine water vapor line absorption and continuum absorption, SARTA relies on a fundamental rule of radiative transfer: **transmittances multiply, while optical depths add.**

SARTA performs separate polynomial calculations for the lines and the continuum, and then combines them at the channel level for every atmospheric layer.

Step 1: Separate Layer Computations

For any given atmospheric layer (L) and instrument channel, SARTA concurrently solves two distinct regression equations:

1. **Line Optical Depth (τ_{line}):** Calculated using the 11-term polynomial set (incorporating the $\sqrt{W_r}$ saturation terms).
2. **Continuum Optical Depth (τ_{cont}):** Calculated using the collision-based polynomial set (incorporating the W_r^2 and layer pressure terms).

Step 2: Combining the Optical Depths (The Additive Step)

Because optical depth is a linear measure of total molecular cross-section, SARTA simply adds the two values together to get the **Total Water Vapor Optical Depth (τ_{H2O})** for that layer:

$$\tau_{\text{H2O}}(L) = \tau_{\text{line}}(L) + \tau_{\text{cont}}(L)$$

If other gases absorb light in that same channel (such as a fixed gas like CO₂ or ozone), their calculated optical depths are added to the sum as well:

$$\tau_{\text{Total}}(L) = \tau_{\text{H2O}}(L) + \tau_{\text{Fixed}}(L) + \tau_{\text{O3}}(L)$$

Step 3: Transformation to Transmittance (The Multiplicative Step)

Once the total optical depth for the layer is established, SARTA applies Beer's Law along the slanted satellite path ($\sec\theta$) to compute the **Layer Transmittance (t_{layer})**.

Mathematically, passing the added optical depths through an exponential function automatically turns the relationship into a multiplication of transmittances:

$$t_{\text{layer}} = \exp(-\tau_{\text{Total}} \cdot \sec\theta) = \exp(-\tau_{\text{line}} \cdot \sec\theta) \times \exp(-\tau_{\text{cont}} \cdot \sec\theta) \times \exp(-\tau_{\text{other}} \cdot \sec\theta)$$

$$t_{\text{layer}} = t_{\text{line}} \times t_{\text{cont}} \times t_{\text{other}}$$

Why this combination matters for Satellite Imagery

This combination directly dictates what a satellite sensor sees:

- **In the Atmospheric Infrared Windows (e.g., 8–12 μm):** Water vapor line absorption (τ_{line}) is very weak, but the continuum (τ_{cont}) is highly active. In humid tropical regions, the W_r^2 continuum term dominates, lowering the total transmittance (t_{layer}) and hiding the true temperature of the Earth's surface.
- **In the Water Vapor Absorption Bands (e.g., 6.3 μm):** The line absorption (τ_{line}) is incredibly strong and saturates high up in the troposphere. The continuum math becomes secondary because t_{line} drops to near zero, completely blocking any thermal radiation from reaching space from the lower layers.

Once SARTA calculates this final combined layer transmittance (t_{layer}) for all 100 levels, it passes the vertical grid directly to the numerical integration engine to build the top-of-atmosphere radiances.

Now that you see how the lines and continuum fuse together, would you like to explore how SARTA updates its **coefficients when new versions of the MT_CKD continuum model** are released, or should we look at how **clouds intercept** this combined thermal stream?